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# Integrated Service Technical Application - Programming (ISTA/P)

**Model: All Models**

**Production: All**

## OBJECTIVES

After completion of this module you will be able to:

- Navigate through the various ISTA/P menus
- Successfully code/program a BMW Group Vehicle
- Understand the reports generated by ISTA/P
- Know how to import a VO
- Know how to import an enabling code
- Know how to update the Navigation maps
- Know how to update the Gracenote® DB

# Introduction

## Reasons for introduction of the new programming system

Due to the constantly growing complexity and ongoing development of functions over the life of vehicles, adaptation of the software on the control units is unavoidable.

The launch of the F01 brought with it a new generation of vehicles which have not only new functions but also a new control unit architecture and a new electrical system configuration.

In order to ensure that those technologies can be supported by the dealer organization today and in the future, a new workshop system was introduced alongside them.

The new workshop system incorporates an extensive workshop network with a central server for data storage and provision of the applications. It also introduces new hardware components for diagnosis and programming.

The existing programming system, Progman, is being replaced by the Integrated Service Technical Application for Programming, ISTA/P.

ISTA/P contains the present vehicle field programming functions plus new functions and is thus designed for the future content and requirements of vehicle programming in the field.

**Note: For more information regarding ISTA/P, please refer to Dealernet and select: Menu>BMW>Aftersales Portal>Service>Workshop Technology and access the ISTA/P User Documentation file. You can also type Workshop Technology in the search engine and that will prompt you to the correct web page.**

---

## **New features when comparing ISTA/P with Progman**

- Reduction and combination of input demands.
- Configurable measures plan viewable as a graphical chart or a table.
- Detailed display of control unit status.
- Automatic repetition of programming in the event of control unit programming or coding errors within the programming process.
- Importing of enabling codes/vehicle order from any location in the workshop provided there is access to the workshop system/ISTA/P.
- Measures plan can be subsequently extended and adapted.
- Programming takes place first followed by automatic coding of all control units.
- Proactive saving of customization and CBS data and re-importing into control units from F01 on.
- Saving of personal profile settings such as phone book or navigation destinations, depending on control units fitted.
- Generation of a control unit order list if hardware needs to be replaced.
- Session can be saved if parts are not immediately available.
- Display of the Progman or ISTA/P version that was last used to program the vehicle.
- Timely/faster updates.

---

## Release Phases

The new programming system ISTA/P is to be introduced in several phases.

### Phase 1

Phase 1 requires the transformation of the Software Service Station, SSS, into the Integrated Software Service Station, ISSS. This “Phase 1”, allows the ISSS to be integrated in the ISIS network.

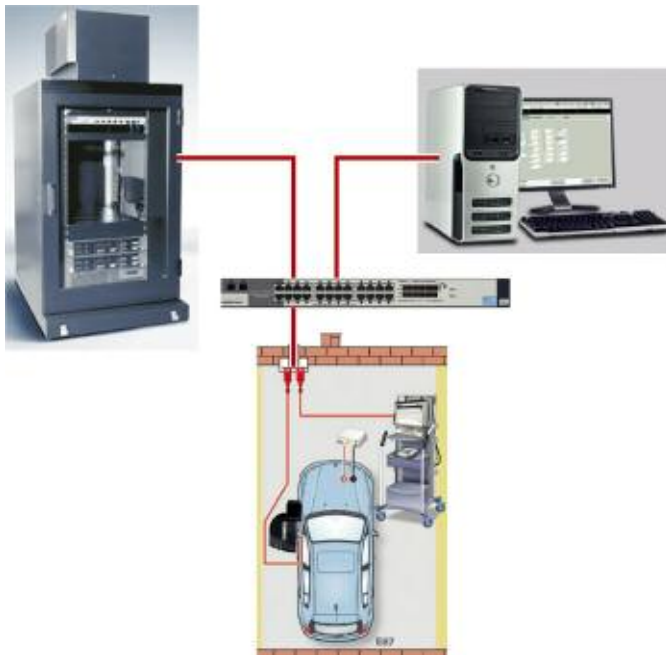
The actual data/software (ISTA/P application) needed to program the vehicles is installed in the will be on the ISSS. Updates will be applied to the ISIS via Jetstream or from DVD and then passed on to the ISSS during the maintenance cycles (off-business hours).

ISTA/P is capable of programming and coding all models.

Initially, the interface and control for programming can only take place by using keyboard and mouse on the ISSS. Soon after, programming control can take place via each individual ISID in the workshop.

The ICOM A and the ICOM B will be used as vehicle interface for programming. The ICOM B will be used exclusively for programming MOST control units.

From the introduction of the ethernet connection in the OBD2 interface, ICOM B will not be used. (example: F01/F02)



**Illustration of Phase 1**

### Future Phases

It is planned to integrate the programming system ISTA/P entirely in the ISIS workshop server. A precise timetable for integration of ISTA/P in ISIS has not yet been established.

---

## Transforming an SSS into an ISSS

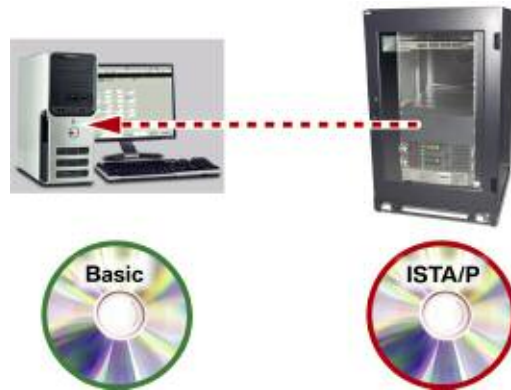
The technical requirements for conversion demand that the SSS is at least the 2nd generation (release 2). The 1st generation SSS can not be converted.

A starter pack containing the following items will be needed:

- ISSS - basic DVD (1 disc)
- ISIS - ISTA/P client (1 disc)
- ISIS ISTA/P data/software (2 disc)

First, the basic DVD is placed in the DVD drive on the SSS and the operating system installed on the SSS. In the process, the previous application, Progman, is completely deleted. From this point on, future, updates and installation of the ISTA/P application on the ISSS will take place exclusively via the ISIS.

The ISTA/P client and then application software are installed on the ISIS. The ISSS is updated/supplied with the application software by the ISIS via the network during the maintenance cycle.



## Programming

An important factor for error-free programming is proper preparation of the vehicle and observance of the instructions during programming. There are special requirements that may apply to individual vehicle models. The actions that are required for all models are described in the following pages.

**Note: Programming must always be preceded by carrying out a diagnosis on the vehicle and making sure that there are no faults on the vehicle. Programming must not be started before faults in the vehicle electrical system are ruled out. Always refer to the latest bulletins and information on currently known issues involving software releases. Please refer to SI B 01 03 06 for further information.**



## Preparatory Measures

|                                                                                                      |                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>TE04-4830</p>   | <p><b>Engine</b></p> <p>Turn off engine, ignition key turned to terminal 0</p>                                                                                                                                                                                             |
|  <p>TE04-4831</p>   | <p><b>Manual Gearbox/SMG/DKG</b></p> <ul style="list-style-type: none"> <li>- Transmission in neutral</li> <li>- Parking brake applied</li> </ul>                                                                                                                          |
|  <p>TE04-4832</p>  | <p><b>Automatic Transmission</b></p> <ul style="list-style-type: none"> <li>- Transmission in position P</li> <li>- System temperature below 80°C</li> </ul>                                                                                                               |
|  <p>TE08-0564</p> | <p><b>Parking Brake</b></p> <p>On vehicles with electromechanical parking brake, apply the parking brake by means of the parking brake button, otherwise use the handbrake.</p>                                                                                            |
|  <p>TE04-4833</p> | <p><b>Electric Loads</b></p> <p>All electric loads, lights and turn indicator switched off.</p> <p>Wiper/washer system switched off. Make sure that the wipers can move freely.</p> <p>The wipers may be activated during programming. On no account block the wipers.</p> |

|                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <p><b>Battery and Battery Charger</b></p> <p>The battery should be sufficiently charged at the start of the programming procedure (&gt;13 V).</p> <p>Connect a BMW-specified and approved battery charger and set to external power supply mode (FSV mode). Do not connect or disconnect the charger during programming. The electrical system voltage must not drop below 13 volts during the programming procedure.</p> <p><b>Note: If Programming/Coding a Hybrid Vehicle please follow their specific preparatory measures.</b></p> |
|    | <p><b>Diagnosis</b></p> <p>Carry out a vehicle test on the ISTA workshop system.</p> <p>Using the ISTA diagnosis system, rectify any problems before programming and delete stored fault codes.</p>                                                                                                                                                                                                                                                                                                                                     |
|   | <p><b>Interfaces</b></p> <p>Programming is performed exclusively via the ICOM devices.</p> <p>Any vehicle equipped with a MOST bus manufactured before the F01/F02, will require the use of an ICOM A and an ICOM B.</p>                                                                                                                                                                                                                                                                                                                |
|  | <p><b>Programming</b></p> <p>CKM data is automatically saved by ISTA/P and written back to the control unit after programming.</p> <p>The data status of the ISSS must always be up to date.</p>                                                                                                                                                                                                                                                                                                                                        |

## Hybrid Vehicles

To properly diagnose E72 Active Hybrid X6 vehicles, all current software must be installed on your ISSS & ISTA/P equipment. Please refer to the Workshop Technology website on Center Net for the update instructions contained in the "ISSS & ISTA/P Installation Manual" and for other relevant information - Release Notes, 3g DCS messages etc. The update process takes approximately 2-3 hours.

Please ensure that no overnight programming happens when the updates are scheduled.

Updated information:

- Please check that the transport / energy saving mode is off.
- All E72 Active Hybrid X6 vehicles need to be on the integration level (I-level) E070-09-12-512 or higher.

- 
- Before start programming, please make sure that the **HV (High Voltage) is switched off** with Service Disconnect.
  - Keep in mind that the car has to go to sleep mode before you can start the engine after programming.
  - The E72 Active Hybrid X6 is fully supported by BMW NA Technical Services.

### **During Programming/encoding**

- Take note of and comply with any prompts or instructions issued by ISTA/P.
- Leave the ignition on and comply with the instructions issued by ISTA/P (e.g. for CAS).
- Do not cut off the connection between the workshop network, ICOM and vehicle.
- After every action in the luggage compartment (e.g. loading / removing navigation DVD), close the luggage compartment lid of the vehicle (prevents the luggage compartment lights from overheating).
- Do not do anything in or on the vehicle during vehicle programming/encoding unless the ISTA/P system has issued an instruction to do so.

### **Follow-up**

- The follow-up work listed at the end of the final report such as calibrations, initializations, service functions, etc. are to be carried in the vehicle test in the ISTA workshop system. Each item of follow-up work is to be confirmed individually.
- Enter time and date information in vehicle via iDrive for correct CBS value calculation.
- After vehicle programming/encoding, insert/connect all data storage media (DVD, iPod®, etc.) that were previously removed/disconnected in the drives and interfaces in the vehicle.
- Compare the previously printed out or noted CBS data of the control units and, if required, reset and correct them via the vehicle test in the ISTA workshop system.
- Run the vehicle test with the ISTA workshop system and terminate the vehicle test.
- For safety reasons, keep vehicle stationary for at least 5 minute (vehicle electrical system 2000 and vehicle electrical system 2020 vehicles) or 16 minutes (I-Bus vehicles) with terminal R OFF so that all control units "go to sleep", i.e. can assume rest state.

**Note: Control units that do not go into "sleep mode" could cause errors relating to closed circuit current!**

- Concluding check that the vehicle is trouble-free.
- Check all radio and TV frequencies and destinations previously noted and store manually as necessary.

---

## Programming of Control Units

The programming system ISTA/P ensures updating of the data statuses on all vehicles by automatically generating a measures plan that should be implemented on vehicles that are bound to an I-level status.

In this context, a distinction is made between three different options. Each of these options represents a change or adaptation to the software and functions. A distinction is made between three measures:

- Programming
- Coding
- Customizing

### Programming

Programming (also known as flash programming) loads a new program or data in the control unit. A distinction can be made between control units with:

- Program status and
- Program and data status

The program status of the control unit corresponds to the operating system and controls the computer program in the control unit. The data status involves the characteristic maps and characteristic curves specific to the vehicle, engine and transmission. The ISTA/P programming application automatically takes that into account when programming the control unit concerned.

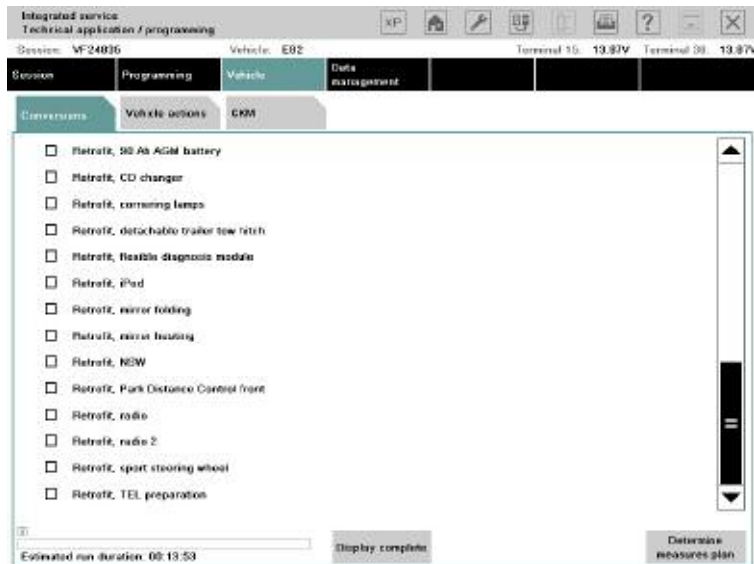
With ISTA/P, the control unit has been programmed, any necessary coding procedures are performed automatically.

### Coding

Coding involves adapting the control units to the specific vehicle. That means that functions and data maps already in the control modules are enabled or activated in accordance with the vehicle order for the vehicle.

## Customizing

On older vehicles, the car and key memory (CKM) option located under the Vehicle option is used to enter the customer-specific settings for the car and key memory functions on certain electrical systems.



## Personal Profile

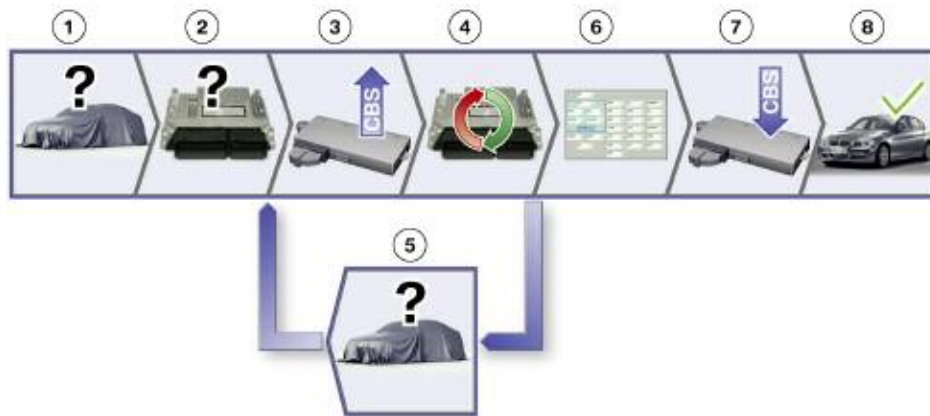
On vehicles with the Personal Profile function, the customization settings are entered directly by the customer. For such vehicles, there is no Customize button on the Vehicle menu.

## Programming Process

The programming process consists of various procedures, some of which require manual intervention although most are automatic.

The individual procedures are illustrated here and described in detail below.

### Programming process sequence



| Index | Explanation                                                         |
|-------|---------------------------------------------------------------------|
| 1     | Identify vehicle and obtain read-out of control unit data           |
| 2     | Produce and configure measures plan                                 |
| 3     | Prepare for programming, export CBS/CKM data                        |
| 4     | Carry out repairs and replace control units where necessary         |
| 5     | Re-check vehicle identification after repairs. Update measures plan |
| 6     | Carry out programming                                               |
| 7     | Carry out programming follow-up, import CBS/CKM data                |
| 8     | Programming successfully completed                                  |

## Starting Programming

Essentially, ISTA/P is an independent external application that will run on the ISSS in Phase 1.

Although in a latter stage ISTA/P can be started from the ISID, it is entirely independent of the ISTA diagnosis application. That means that before programming can be started, diagnosis must be ended or interrupted.

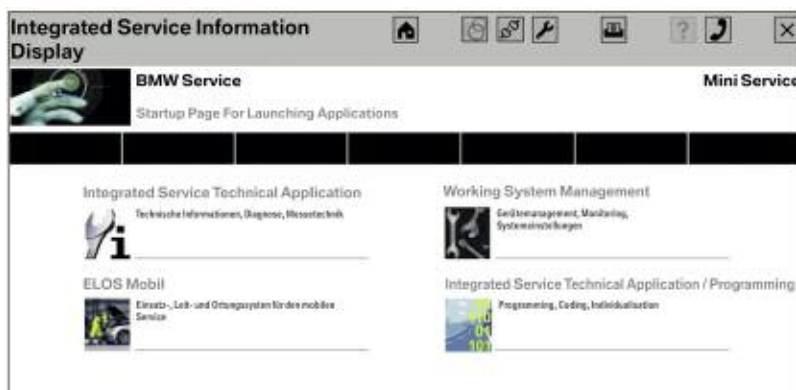
Starting and management of a new programming session can only be carried out by means of ISTA/P.

When ISTA/P is started, the user has a choice of three different actions:

- Cancel and exit ISTA/P
- Open an existing programming session
- Start a new programming session



**Start screen on the ISSS Integrated Software Service Station**



**Start screen for ISID (not enabled initially)**

## Opening an Existing Programming Session

If the user decides to open an existing programming session, all the programming sessions currently in progress are listed and the user can select the session required.

On the toolbar there are various buttons for opening more menus.

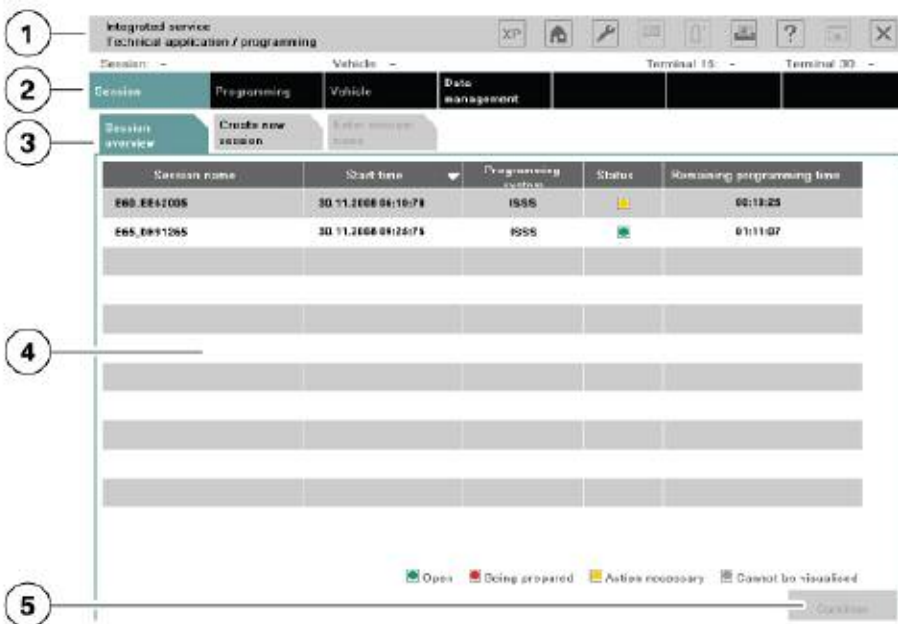
On the options bar, you can select the session or the vehicle that is to be programmed or the Integrated Software Service Station (ISSS) that is to be used for programming. Depending on the model of the vehicle, each ISSS can carry out up to five programming sessions at the same time.

The button for selecting the vehicle access method is located on the function bar.

The information panel shows the details of the sessions or available ICOMs.

The control buttons are used to navigate through the programming process, e.g. the "Next" button takes you to the next screen.

### List of current sessions



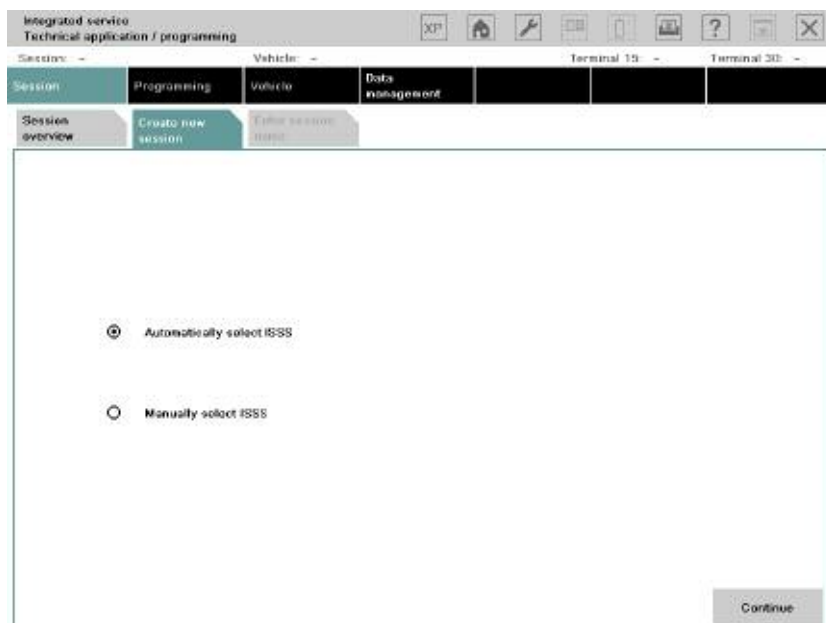
| Index | Explanation      |
|-------|------------------|
| 1     | Toolbar          |
| 2     | Options bar      |
| 3     | Function bar     |
| 4     | Information pane |
| 5     | Control buttons  |



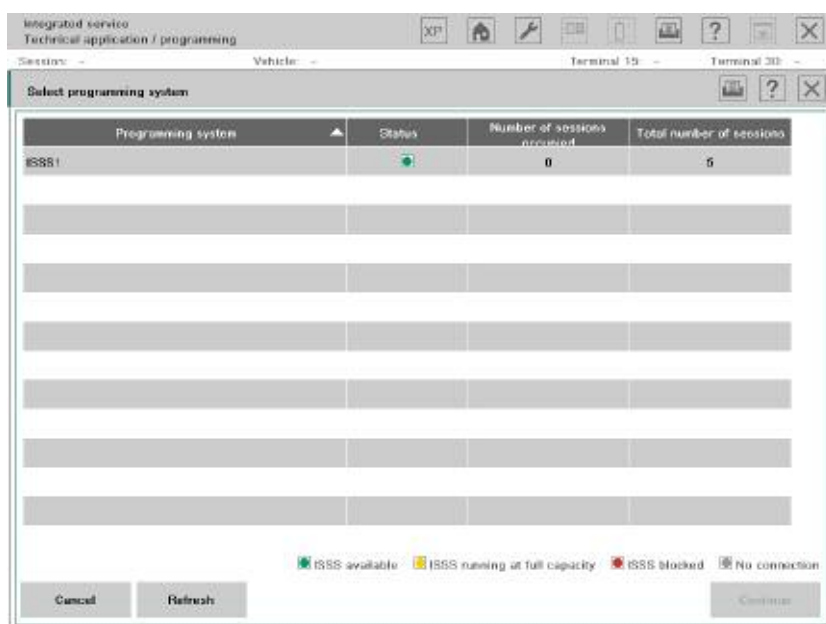
## Starting a New Programming Session

When a new programming session is to be started, the programming server on which the vehicle is to be programmed must first be selected.

The ISSS used for the session can be selected manually or automatically. Because at the moment the ISSS can only program one F01 at a time or 5 other vehicles at a time, it is recommended that the automatic ISSS selection is carried out.

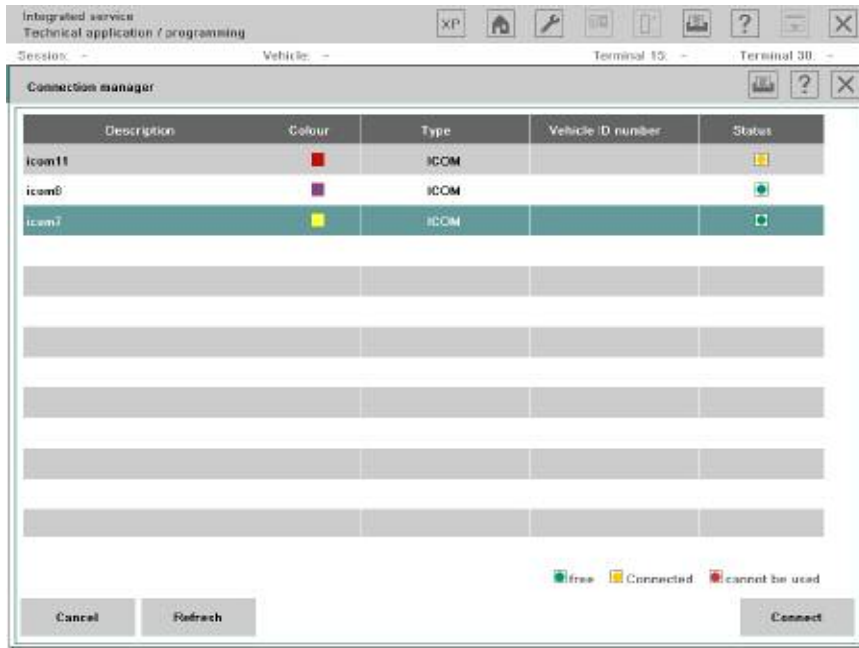


### Selecting ISSS to be used for session



### Selecting ISSS to be used for session

The connection manager on the ISIS establishes all the ICOMs that are available and displays the details. The user can select the desired ICOM or vehicle from the list displayed. ISTA/P then establishes the vehicle data. If that is not possible for any reason, the vehicle must be selected manually.



### Selecting ICOM

The user then has the opportunity to give the session any individual name, but the system automatically enters the model of the vehicle along with the chassis number as an identifier (example: E60\_EE12345).

---

## Selecting Vehicle Interface

The same vehicle interfaces are used for programming with the new workshop system as for diagnosis.

From the F01 on, all control units can be programmed using only the **ICOM A**. Even though the F01/F02 is equipped with a MOST bus, **ICOM B** is not necessary due to the ethernet connection via the OBD2 connector.

For CAN-bus vehicles the ICOM A is used. MOST control units are programmed directly using the ICOM B. The ICOM B is connected to the ICOM A and the workshop network by a USB cable.

For older I-bus/K-bus vehicles with the round diagnosis connector, the ICOM C is used in conjunction with the ICOM A.



**Vehicle interfaces**

---

### **Vehicle Access for MOST Bus**

When programming vehicles with MOST bus from E65 to E9X, vehicle access is established as shown in the photograph below.



**Vehicle access for programming MOST-bus. The location of the MOST connector varies according to the vehicle.**

### **Vehicle Access for I-bus/K-bus Vehicles**

On older vehicles without OBD II connection, programming is carried out via the round diagnosis connector in the engine compartment. That requires connecting the ICOM C to the ICOM A and on to the workshop network.



**Vehicle access for I-bus/K-bus vehicles**

### **Vehicle Access from the F01/F02 on**

All control units will be programmed exclusively using the ICOM A.

---

## **Identifying the vehicle and obtaining a read-out of control unit data**

ISTA/P checks the vehicle electrical system, the central gateway and the vehicle-order control units (FRM/CAS/light module). The vehicle order (VO) is read and a consistency check carried out between the actual status and the required status.

If the central gateway does not respond, a software routine for restoring it to working order is attempted before the service technician is instructed to replace the ZGW.

First of all, the basic details of the vehicle are established such as VIN number, model, type code, etc.

Then, all control units fitted and the current I-level of the vehicle is determined. In addition, the programming system establishes, if possible, the software version last used to carry out servicing operations on the vehicle.

That makes it possible to avoid unnecessary programming if the vehicle already has the latest software version.

Afterwards, the details of the ex-works equipment options for the vehicle stored on the BMW programming system are retrieved.

Next, the control units identified and the equipment options fitted are compared (comparison of required vs. actual status).

For example, it is established whether the list of control units obtained actually matches those fitted on the vehicle and whether the level matches the current required status.

That ensures that all control units on the vehicle communicate properly with one another within the network. An I-level is a combination of networked control units validated and approved by BMW Development.

## Establishing vehicle details

Integrated service  
Technical application / programming

Session: VF24836 Vehicle: EE2 Terminal 15: 12.43V Terminal 30: 12.43V

Session Programming Vehicle Data management

Vehicle details

Vehicle ID number: VF24836 E designation: EE2  
Mileage: 401 km Type/Description: EE2 Sedan M47020 RHD ECE  
Type code: UC73 Model year: 2008  
Paint code: 6A72 Upstream code: LWSW  
Integr. level (plant): E892-08-03-520 Integr. level (actual): E892-08-03-515

Last treated with: Program 21.1

Optional extras:

| Number | Description                               |
|--------|-------------------------------------------|
| 205    | Automatic transmission                    |
| 217    | Active steering                           |
| 249    | Multifunction for steering wheel          |
| 25W    | Light alloy wheels star spoke 204 with... |
| 2v9    | Tire Pressure Monitor                     |
| 319    | Integrated universal remote control       |
| 321    | Exterior parts in body colour             |
| 322    | Comfort Access                            |

Activation code status (before plan execution):

| Abbreviation | Application number | Upgrade | Status |
|--------------|--------------------|---------|--------|
|--------------|--------------------|---------|--------|

Determining vehicle features

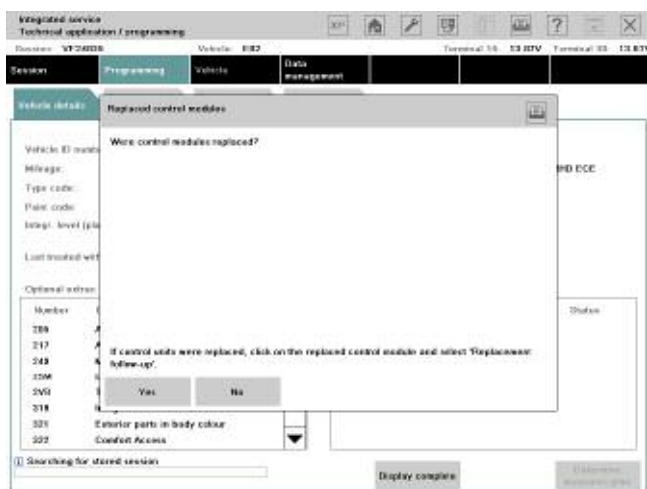
Display complete

| Index | Explanation                                          |
|-------|------------------------------------------------------|
| 1     | Vehicle data                                         |
| 2     | I-level with which the vehicle left the factory      |
| 3     | Current I-level of the vehicle                       |
| 4     | System and version last used to program the vehicle  |
| 5     | List of all optional equipment fitted on the vehicle |

## Producing and Configuring the Measures Plan

Before programming can be carried out, the measures plan is produced and can also be configured by the user. For example, it is now possible to program retrofits or conversions at the same time as the software update. The configuration for that is done now so that no further manual input is required later on during the programming sequence and it can then run automatically.

After the identification process, the user is asked if any control units have been replaced on the vehicle. If the answer is YES, ISTA/P checks whether an enabling code is required for the replaced control unit and, if so, requests it from the user if it has not already been entered on the system.



### Control unit replacement check

Associated customization data such as settings for the air conditioning are backed up and the action "Follow up control unit replacement" added to the list of actions.

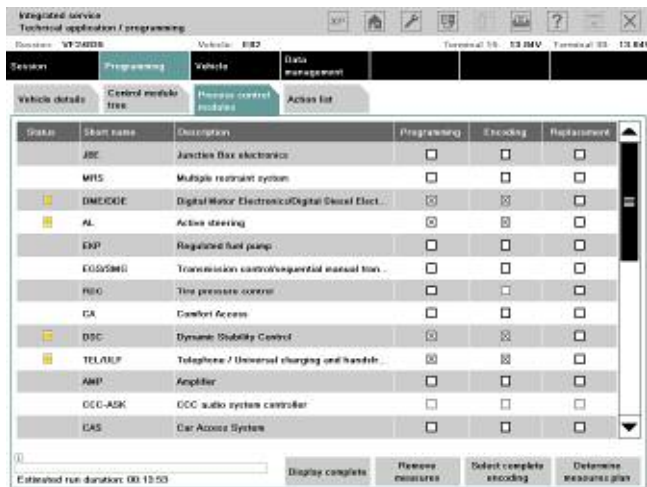
In general it is advisable to manually back up the customer's individual settings. If the control unit is defective, they can not be read and, therefore, can not be written back to the control unit after programming either.

The necessary actions, such as "Update control unit software", are established and added to the list of actions.

Then for the first time ISTA/P produces a measures plan based on the tasks in the list of actions at that point.

The user has the option of editing the measures plan. The user can add further actions to the measures plan but not remove them. The following actions can be added to the measures plan:

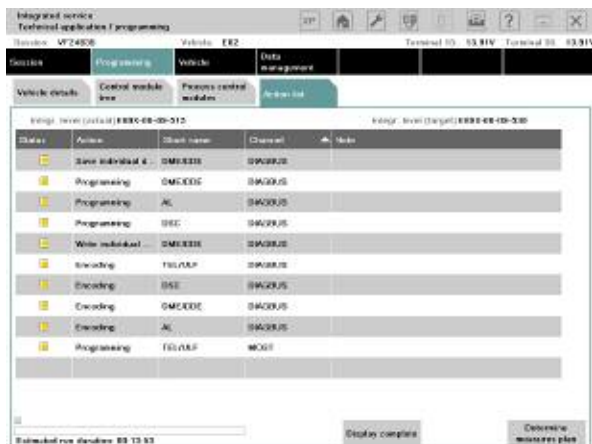
- Carry out conversion
- Prepare control unit replacement
- Program control unit
- Code control unit
- Set CKM data
- Enter enabling code
- Import vehicle order



**Note:** Required service functions such as initializations and clearing of fault codes are only indicated; they must be carried out on the ISTA diagnosis system.

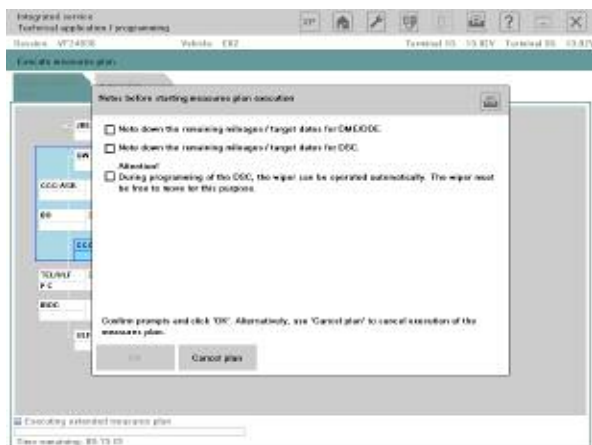






## Action list

The Action list contains all information relating to the concerned control unit and which via channel (D-CAN/MOST) it is programmed.



## Guidance notes

Furthermore, additional guidance notes can be shown, e.g. on noting the remaining life of DME or conversion instructions for replacing control units.

**Note: The guidance notes must be acknowledged by ticking the checkboxes and confirmed by clicking OK.**

---

## Preparing for Programming

After confirming the measures plan, the flash programmability of the control units is first checked and a notification message displayed if there is a problem. Then the customization data and the CBS data is read and backed up for rewriting later on.

If a control unit requires replacing, ISTA/P displays a message to the user indicating that the control unit should now be replaced.

After replacement is confirmed, the action "Follow up control unit replacement" is flagged for attention later on in the process.

ISTA/P then shows all required notifications, which must be acknowledged by the user before the actual programming starts.

That simultaneously starts the automatic programming process which requires no further interaction on the part of the user.

**Note: As always, please keep in mind that the CAS is programmed first with the key removed. After successfully programmed, the ignition has to be switched on again.**

## Carrying Out Repair Tasks

If it was established in the process of identifying the vehicle that repair tasks are required, e.g. due to updating hardware or replacing a control unit, they are now carried out before the actual programming/coding process.

If the spare part is not available or the wrong part has been ordered, the session can be saved and ended at this point. As soon as the right part is available, the session can be retrieved and continued.

ISTA/P establishes from the measures plan which repair tasks are required and displays them. After successful completion of a repair task, the vehicle must be re-identified and any necessary software updates added to the measures plan.

In that way the possibility of inconsistencies on the vehicle after a repair is avoided. If a repair task fails, the programming process is cancelled.

## Updating the Measures Plan

After a repair task has been carried out, ISTA/P performs the vehicle identification process again. If it identifies irregularities, e.g. that a control unit with an older I-level status has been fitted, appropriate actions are automatically set on the system or indicated to the user.

If there are irregularities, ISTA/P establishes which software updates are required based on compatibility management and adds them to the action plan. The measures plan is then updated and programming automatically started.

To prevent vehicles with inconsistencies being created, the user has no means of stopping the programming sequence.

## Carrying Out Programming

This stage of the process involves the programming and subsequent coding of the control units.

In the process of programming or coding a particular control unit, the system checks in each case whether the operation has succeeded before continuing with the programming of further control units. A particular version of ISTA/P distributes only one I-level. Programming of the vehicle as a whole (meaning all control units on the vehicle) is always performed to the latest I-level (target I-level).

The next step is to write the enabling codes to the relevant control units if required and modifying the vehicle order if conversions have been carried out. The programming of control units can also be performed simultaneously if they are on different bus systems, e.g. MOST and CAN.

The progress of the programming process is indicated by a progress bar.



**Carrying out programming**



**View after completion of programming**

If programming/coding/initialization has been completed successfully, there is a green tick by every control unit on which an action has been performed.

If an action has failed or not been completed, the control unit concerned is marked with a red X.

By clicking the control unit concerned, you can obtain information as to why programming failed.



## Completing Programming Session

The last stage of the programming process is importing the customization, CBS and CKM data. The data is written back to the control units and checked. Afterwards, the automated initialization sequences are performed where required (electric windows/slide/tilt sunroof).

The automatic programming process is then complete. The initialization sequences that require user interaction are then carried out.



### Notes on follow-up

After completion of the initialization sequences, ISTA/P establishes from the measures plan what service functions are required, e.g. setting steering angle. They are shown to the user. They have to be carried out as part of the diagnosis after programming. That is carried out by way of the vehicle test on the ISTA workshop system and afterwards all fault memories must be cleared.

Finally, the actual status is checked once again and the final report produced. ISTA/P provides the user with the final report.

The user can print out the final report or else it is stored for a certain period on the ISIS.

The entire programming log is sent to BMW AG as part of the FASTA data. This allows BMW to retrace how the vehicle programming was carried out from the user actions, either for accounting purposes or if there is a subsequent fault.

Furthermore, that data is automatically analysed so that software errors can be detected at an early stage and suitable measures initiated. As a result, BMW is able to continuously monitor and improve software quality.

## Final Report

### 1 Final report

Session name: ES2\_VF24628  
 Total duration: 00:28:13  
 Concluding work, notes: present (see below)  
 Change vehicle order: No

#### Header data

2 Date: 11/15/2018 Time: 11:12:00 AM ESTAP version: 31.24.8

3 Model entry: 602 Type/production: VCTD / 507 Sedan (HTD2) PHD 50E

4 Mileage: 431 Vehicle ID number: VSAUCT3875F24828

5 Integration level (plant): Integration level (site): Integration level (pack):

6 020V-00-00-000 020V-00-00-015 020V-00-00-000

7 Production date: 0000 Part number: 3470 Updatable code: LANCW

8 E words: 0000

9 K words:

10 SA numbers: 236 217 240 226 210 210 221 522 347 403 400 401 441 442 482 483 473 481 488  
 494 492 494 502 507 521 522 524 524 540 562 570 504 600 622 625 645 652 686  
 677 694 695 697 698 704 710 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999

### 6 Programmings sequential OBD (K line, DCM)B

Result coding according to execution sequence

| Control module | Action      | Result / fault code | Status | Type | Hardware index | CM part number | New part number |
|----------------|-------------|---------------------|--------|------|----------------|----------------|-----------------|
| CAS            | Programming | Successful          | ✓      | VAN  | C4             | -              | 9147227         |
| DME DDE        | Programming | Successful          | ✓      | Sys  | 60             | 7994021        | 7994020         |
| AL             | Programming | Successful          | ✓      | Sys  | 63             | 6791167        | 6791160         |
| DSC            | Programming | Successful          | ✓      | Sys  | 62             | 6791166        | 6791164         |

### 7 Programmed control MOST

Result coding according to execution sequence

| Control module | Action      | Result / fault code | Status | Type | Hardware index | CM part number | New part number |
|----------------|-------------|---------------------|--------|------|----------------|----------------|-----------------|
| TRUSLP         | Programming | Successful          | ✓      | Sys  | 18             | 9199978        | 9199979         |

### 8 Encoding

Result coding according to execution sequence

| Control module | Action   | Result / fault code | Status | Type | Hardware index | CM part number | New part number |
|----------------|----------|---------------------|--------|------|----------------|----------------|-----------------|
| CAS            | Encoding | Successful          | ✓      | Sys  | C4             | -              | 9147227         |
| TRUSLP         | Encoding | Successful          | ✓      | Sys  | 18             | 9199978        | 9199979         |
| DSC            | Encoding | Successful          | ✓      | Sys  | 62             | 6791166        | 6791164         |
| DME DDE        | Encoding | Successful          | ✓      | Sys  | 60             | 7994021        | 7994020         |
| AL             | Encoding | Successful          | ✓      | Sys  | 63             | 6791167        | 6791160         |

### 9 Data recovery

| Control module | Action                | Result     | Status | Type |
|----------------|-----------------------|------------|--------|------|
| DME DDE        | Save individual data  | Successful | ✓      | Sys  |
| DO             | Save individual data  | Successful | ✓      | Sys  |
| DME DDE        | Write individual data | Successful | ✓      | Sys  |
| DO             | Write individual data | Successful | ✓      | Sys  |

### 10 Additional actions

| Control module | Action                   | Result / fault code | Status |
|----------------|--------------------------|---------------------|--------|
| -              | Vehicle order alignment  | Successful          | ✓      |
| -              | Update integration level | Successful          | ✓      |

#### Concluding work, notes

### 11

1. - (Report) / After completion of programming / encoding, reset the adaptation values of the DME/DOE via the vehicle test in the ISTA software system. Compare remaining privileges / target data in DME/DOE with the information data. If successful, start and complete the remaining privileges / target data.
2. - (Report) / After completion of programming / encoding, commission / adjust the AL via the vehicle test in the ISTA software system.

| Index | Explanation                                                                     |
|-------|---------------------------------------------------------------------------------|
| 1     | Final report                                                                    |
| 2     | Current programming data: data, time, ISTA/P version                            |
| 3     | Vehicle data: mileage and VIN number                                            |
| 4     | I-levels: factory, last, current                                                |
| 5     | SA numbers                                                                      |
| 6     | Programming actions indicating success or failure over MOST                     |
| 7     | Programming actions indicating success or failure over I/K bus                  |
| 8     | Coding actions indicating success or failure                                    |
| 9     | Data saving actions indicating success or failure                               |
| 10    | Additional actions, e.g. whether updating of I-levels was successful or failed. |
| 11    | End notes                                                                       |

# Overview of Functions on ISTA/P

The following graphic in the "Programming" menu, "Edit control units" tab, shows a layout example of a typical screen mask.



| Index | Screen element | Index | Screen element |
|-------|----------------|-------|----------------|
| 1     | Menu           | 3     | Functions      |
| 2     | Tab            |       |                |

## Overview of Main Functions

The functions in ISTA/P can be accessed via certain paths. The following overviews show the menu in which the required function can be executed. Manual selections and information are normally not shown.

| Menu     | Session<br>Is displayed after "Start ISTA/P"        |                                                                                                                                                                                 |                    |
|----------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
|          | Session Overview                                    | Create New Session                                                                                                                                                              | Enter Session name |
| Function | Programming sessions that are running are displayed | Select ISSS automatically/manually<br>Select programming system<br>Control unit information<br>Connection manager<br>Select vehicle/interface, transfer to the Programming menu | Issue session name |

| Menu     | Programming<br>Is displayed after "Start ISTA/P" (manual switch to "Vehicle" menu and "Data management" menu possible) |                                                                                                                                                                                                                                              |                                                                                            |
|----------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
|          | Vehicle details                                                                                                        | Control unit tree                                                                                                                                                                                                                            | Action List                                                                                |
| Function | Display of vehicle details<br>Determine action plan, transfer to "Action Plan" menu                                    | Control unit actions (programming, encoding, replacing, replacement follow-up)<br>Action list (after control unit action)<br>Control unit information<br>Overall encoding<br>Determine action plan, transfer to the Display action plan menu | Planned actions are shown<br>Determine action plan, display transfer to "Action Plan" menu |



| Menu     | Vehicle<br>Is displayed after "Create new session" |                             |                                                                                                                                                                                |
|----------|----------------------------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Conversions                                        | CKM                         | Vehicle actions                                                                                                                                                                |
| Function | Retrofits<br>Conversions                           | Car and key memory settings | Updating of map data, navigation system (HDD update)<br>Enabling of map data for navigation system<br>Update Gracenote® DB<br>Import vehicle order<br>Select complete encoding |

| Menu     | Data management<br>Is displayed after "Start ISTA/P" and after "Create new session" |                                                       |                                        |
|----------|-------------------------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------|
|          | Import/delete enabling code (summarized)                                            | Import/delete vehicle order (summarized)              | Reports                                |
| Function | Import enabling code<br>Enter short enabling code<br>Delete imported enabling code  | Import vehicle order<br>Delete imported vehicle order | Reports of previous sessions are shown |

| Menu     | <b>Display action plan (1/2)</b><br>Is displayed after "Determine action plan" |                                                                                                                     |                                                                          |
|----------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Tab      | Action plan                                                                    | Control Unit Tree                                                                                                   | Action List                                                              |
| Function | Action plan is shown                                                           | Planned actions are shown<br>Action List<br>Control Unit Information<br>Accept action plan, action plan is executed | Planned actions are shown<br>Accept action plan, action plan is executed |

| Menu     | <b>Display action plan (2/2)</b><br>Is displayed after "Determine action plan" |                                   |  |
|----------|--------------------------------------------------------------------------------|-----------------------------------|--|
| Tab      | Order list                                                                     | Activation code list              |  |
| Function | Control units to be replaced are shown together with the order number          | Imported enabling codes are shown |  |

---

| Menu     | <b>Action plan execution complete</b><br>Is displayed after "Accept action plan" |                                                                                                                    |                                                                            |
|----------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Tab      | Final Report                                                                     | Control Unit Tree                                                                                                  | Action List                                                                |
| Function | Final report is shown<br>End programming switch<br>to Session menu               | Executed actions are shown<br>Action List<br>Control Unit Information<br>End programming switch to<br>Session menu | Executed actions are<br>shown<br>End programming switch<br>to Session menu |

---

## Import Vehicle Order (VO)

In order to import the VO of a vehicle, you should go to the Data Management Menu and then click the Import vehicle order tab.

The following data is necessary to generate the vehicle order:

- Which conversions/retrofits are to be carried out, or which fault has occurred;
- 7-character vehicle identification number;
- Dealer number.

**Note: The vehicle order for conversions and retrofits is ordered through the ASAP portal. To eliminate errors, the vehicle order can be ordered through the subsidiary or the regional office.**

**The vehicle order is downloaded in the form of a compressed XML-file (ZIP file). The ZIP file must be unpacked and filed on the root directory of the data carrier (e.g. F:\)**

**The diversity of USB storage media on the market means that it can not be guaranteed that every USB data storage medium will work with ISTA/P.**

All the vehicle identification numbers where the vehicle orders are in the data stock of ISTA/P are displayed in a selection list. The user can select a vehicle identification number and have the corresponding vehicle order displayed. The user can also search for a vehicle identification number by changing the sorting function.

| User action                                                                                                                                                                  | Result                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 1. Select "Import vehicle order" tab.                                                                                                                                        | The vehicle identification numbers for which vehicle orders or central encoding keys are available are displayed. |
| 2. Press the "Import" button to acknowledge.                                                                                                                                 | A prompt to insert the data carrier appears.                                                                      |
| 3. Insert a data medium in ISSS or set up a connection between ISSS and the USB data storage medium.<br>4. Select vehicle order.<br>5. Press the "OK" button to acknowledge. | The vehicle order is imported.                                                                                    |

---

## How to Import a VO?

Prerequisites:

- Vehicle Order (VO) as XML file received via PuMA case or email, e.g., FZ38042.xml
- Media to copy the XML file on:
  - USB stick (recommended)
  - CD-ROM media and CD/DVD burner
  - Floppy disc
- IBAC code provided by RTE/NTE via PuMA case or email,  
e.g., 3R4SMT4NRI65QBI

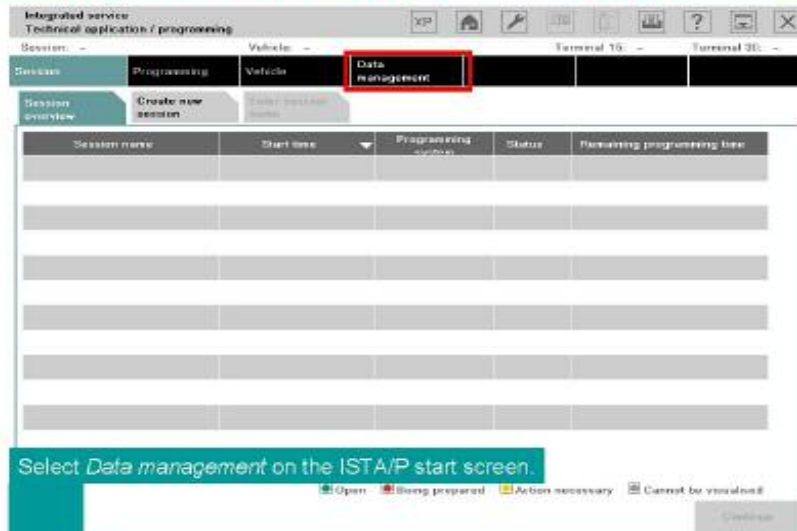
**Note: 1. When using a USB stick, ensure it's formatted as a removable disc (usually default format). Otherwise the stick will not be recognized by ISTA/P.**

**2. More than one VO can be loaded on the interchangeable data media. VOs then need to be separated by different file names.**

**3. Recommended filename is: [7-digit VIN].xml**

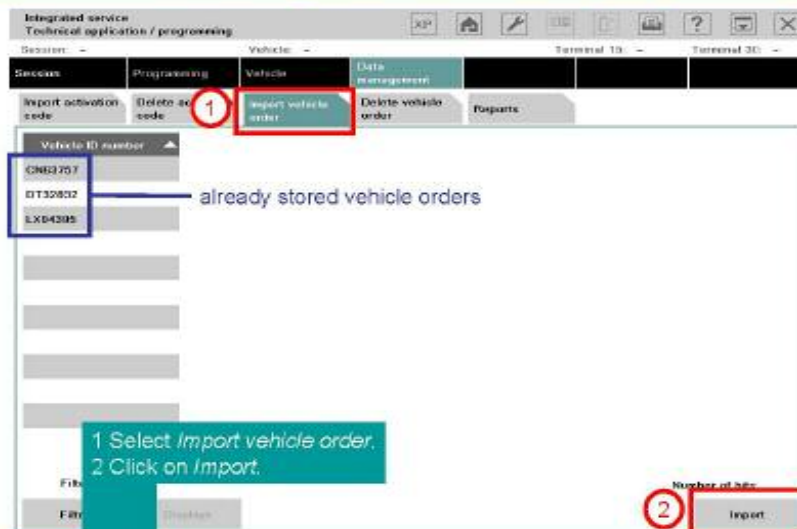
**4. Currently the import of a VO is protected by IBAC. Make sure you have the 15 digit IBAC code before you connect the vehicle.**

## ISTA/P – How to import a vehicle order (VO)



1

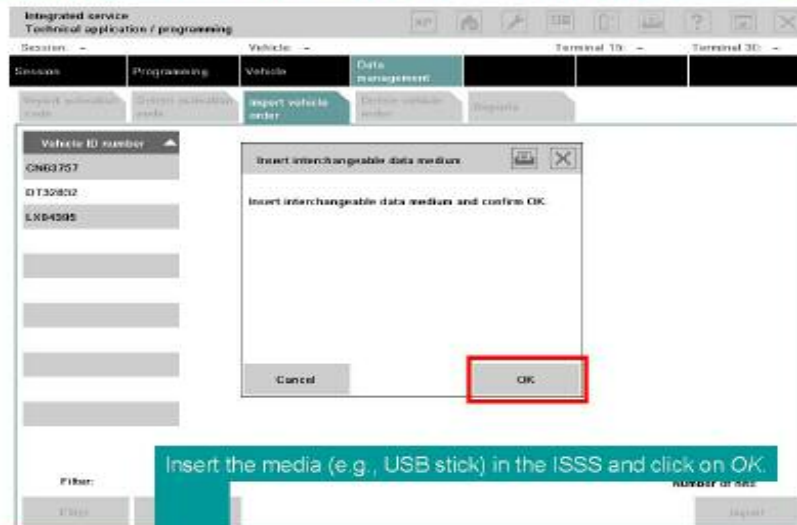
## ISTA/P – How to import a vehicle order (VO)



1

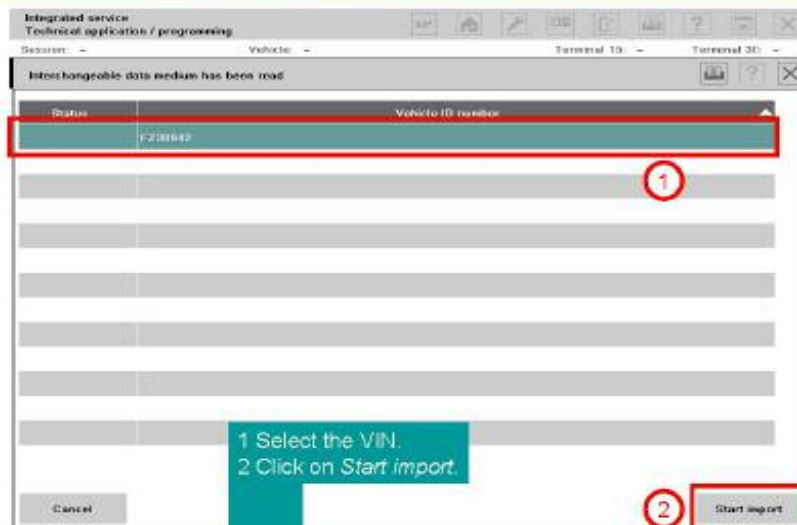
2

## ISTA/P – How to import a vehicle order (VO)



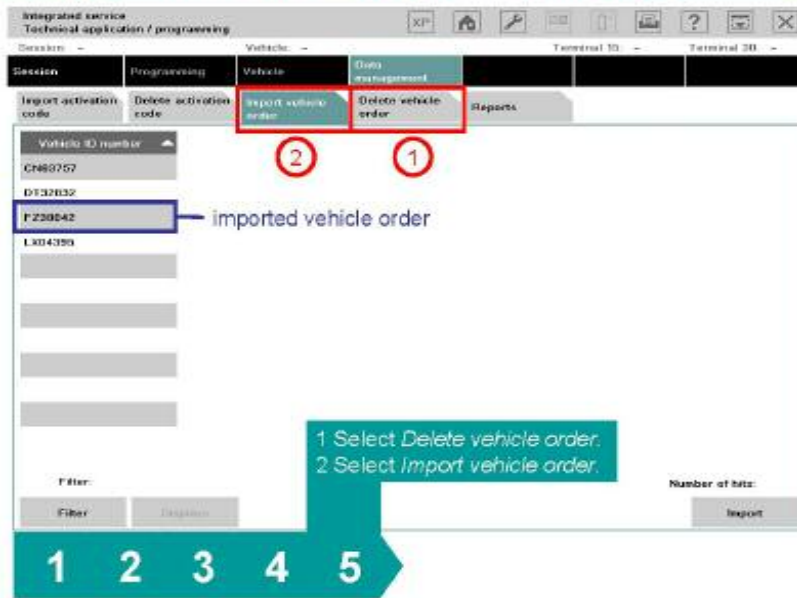
1 2 3

## ISTA/P – How to import a vehicle order (VO)

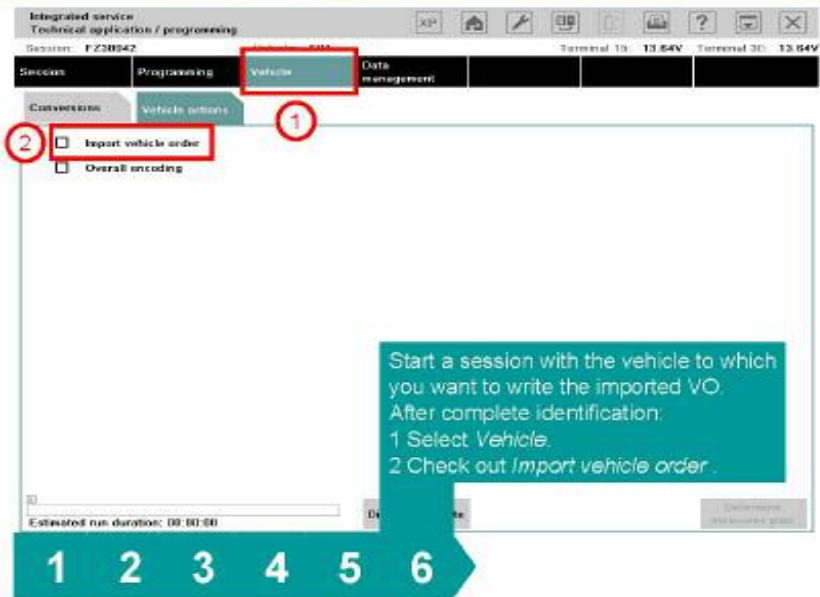


1 2 3 4

## ISTA/P – How to import a vehicle order (VO)



## ISTA/P – How to import a vehicle order (VO)





## ISTA/P – How to import a vehicle order (VO)

Integrated service  
Technical application / programming

Session: FZ38042 Vehicle: E81 Terminal 15: 13.67V Terminal 35: 13.67V

Session Programming Vehicle Data management

Conversions

Vehicle order import

Vehicle order for vehicle identification number W5AVT795XBF-239842 to be written from data management to the connected vehicle?

OK Termination

Updating measures plan  
Estimated run duration: 00:00:00

1 2 3 4 5 6 7

Compare the VINs.  
If they match, click on OK.  
If not, click on Termination.

## ISTA/P – How to import a vehicle order (VO)

Integrated service  
Technical application / programming

Session: FZ38042 Vehicle: E81 Terminal 15: 13.77V Terminal 35: 13.77V

Session Programming Vehicle Data management

Conversions

Input IBAC activation code

IBAC service: Import vehicle order  
IBAC action: Activate  
Folder code: 8045  
IBAC activation code

1 3R4SMT4MRK5Q2B

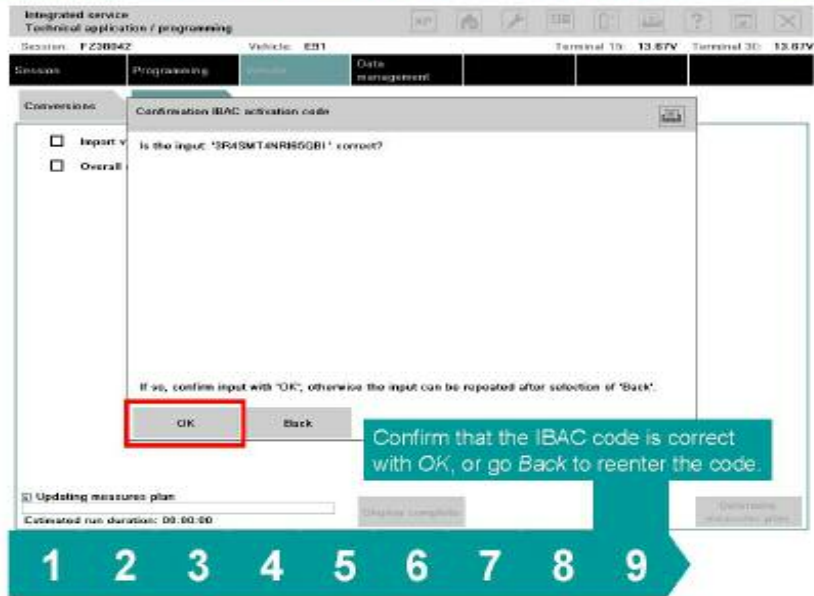
2 OK

Updating measures plan  
Estimated run duration: 00:00:00

1 2 3 4 5 6 7 8

1 Type in the 15 digit IBAC code.  
2 Click on OK.

## ISTA/P – How to import a vehicle order (VO)



## ISTA/P – How to import a vehicle order (VO)



---

## Import Enabling Code

Software enabling in a number of control units (e.g. when programming the CCC, an enabling code must be imported to enable the option "extended voice control") is possible with ISTA/P.

In ISTA/P, there is the possibility to import enabling codes for a vehicle even before working through the action plan. All imported enabling codes are saved within the ISPI network and are available for future sessions without having to import them again.

| User action                                                                                                                                      | Result                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 1. Select "Import enabling code" tab.                                                                                                            | The vehicle identification numbers for which enabling codes are available are displayed. |
| 2. Press the "Import" button to acknowledge.                                                                                                     | A prompt to insert the data carrier appears.                                             |
| 3. Insert a data medium in ISSS or set up a connection between ISSS and the USB data storage medium.<br>4. Press the "OK" button to acknowledge. | The enabling code is imported.                                                           |

**Note: The enabling code can be downloaded in the ASAP portal. The enabling code is supplied on CD with new control units.**

**The ZIP file must be unpacked and filed on the root directory of the data carrier (e.g. F:\).**

**The diversity of USB storage media on the market means that it can not be guaranteed that every USB data storage medium will work with ISTA/P.**

---

## How to Import an Enabling Code

### Prerequisites:

ZIP archive file with:

- release code (FSC) file (.xml) e.g., FZ38042\_00020001.xml
- certificate file (.der) e.g., FZ38042\_00020001.der

Media to copy the files on:

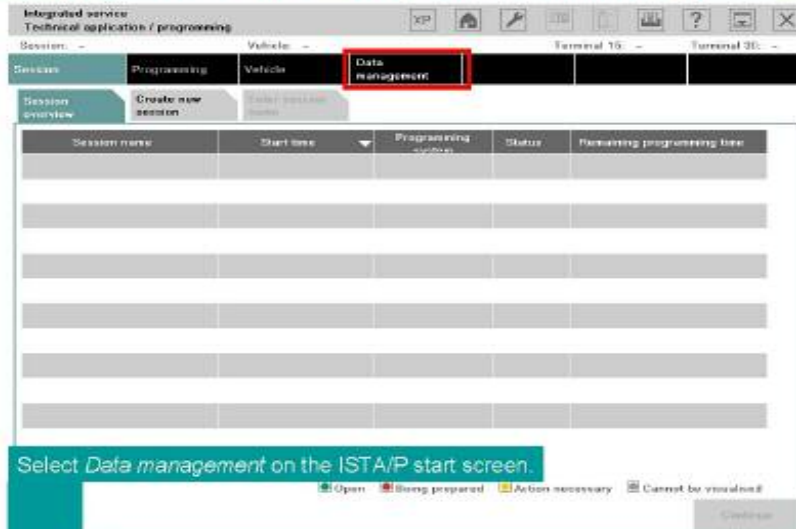
- USB stick (recommended)
- CD-ROM media and CD/DVD burner
- Floppy disc

**Note: 1. This import function neither writes the code to the vehicle nor activates the associated function in the vehicle. The activation will be requested during the measures plan realization.**

**2. When using a USB stick ensure, that it's formatted as a removable disc (usually default format). Otherwise the stick will not be recognized by ISTA/P.**

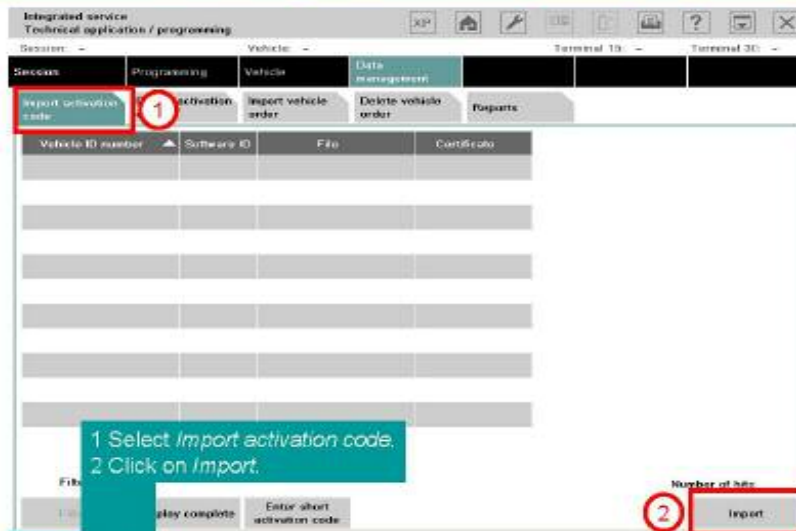
**3. The FSC file (.xml) AND the certificate file (.der) need to be extracted on the media. Do not copy the ZIP archive file on the media.**

## ISTA/P – How to import an enabling code (FSC)



1

## ISTA/P – How to import an enabling code (FSC)



1

2



## Jen



---

## Updating and enabling of map data for navigation system and Gracernote® DB

Updates to map data using ISTA/P are possible only in the case of vehicles with the Car Information Computer (CIC).

Importing of a current navigation map to the vehicle becomes necessary:

- In the event of updating or initialization of the data for new, used and demonstration cars prior to delivery to customer.
- If update is requested by customer.
- In the course of a repair.

The procedure for updating the Gracernote® DB and the map data is identical. Updating of the Gracernote® DB is only possible with the Professional (SA 609) navigation system.

Enabling of the navigation map (Road Map) with ISTA/P is necessary:

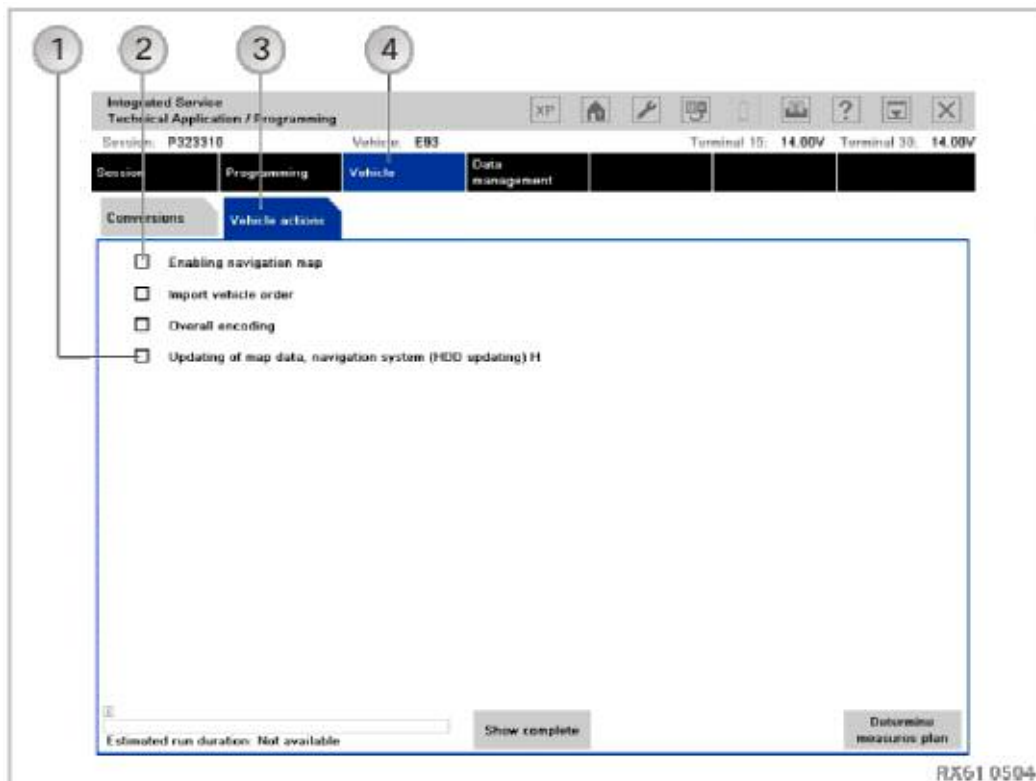
- After updating or initialization of the map data.
- In case of a repair.
- In case of retrofitting of CIC.

**Note:** In the course of a repair, it is not necessary to order an enabling code. In this case, the previous enabling code is used again.

**Updating of the map data for the navigation system (HDD update) depends on the integration levels.**

Proceed as follows to update the navigation system map data (HDD update) or Gracenote® DB:

- To create a new ISTA/P session, see "Vehicle" menu (create new session)
- Select the "Vehicle" menu
- Select the "Vehicle actions" tab



| Index | Screen element                                                                  | Index | Screen element        |
|-------|---------------------------------------------------------------------------------|-------|-----------------------|
| 1     | Checkbox "Update of the navigation system map data (HDD update)"                | 3     | "Vehicle actions" tab |
| 2     | Activate checkbox "Enabling of navigation map (Road Map)", already imported map | 4     | "Vehicle" menu        |

- Activate checkbox "Update of the navigation system map data (HDD update)"

To select further actions with E-series (programming, encoding), switch back to the "Programming" menu.



---

With the F-series, all the previously selected actions will be deleted. No further actions can be added.

- Press the "Determine action plan" button to acknowledge.

The action plan is determined and displayed. The installed navigation maps (Road Maps) and Gracenote® DB are displayed on the ISIS.

**Note: The ISIS server address (ISIS1 or ISIS2) on which the navigation maps with the web copy console are saved must be entered in the "Download-ISIS IP" input box. It is thus identical to the ISIS IP address, which was used by the web copy console.**

**The ISIS IP can be read out in the WSM under device settings.**

**The map data and the Gracenote® DB cannot be updated at the same time.**

**Ensure that the selected Road Map is installed on the ISIS.**

- Enter the Download ISIS IP (e.g. 10.249.155.10) in the input box.

Select Gracenote® DB or "Road Map" by activating the corresponding checkbox

- Press the "OK" button to acknowledge

The name of the Gracenote® DB or Road Map is displayed again for checking purposes.

- Check name
- If the name is OK, press the "OK" button to acknowledge.

The action plan is drawn up.

- Accept the action plan.

After the action plan has been completed, the final report will be displayed.

**Note: After the import, the map data has to be activated. An enabling code is required to do this. To activate the map data, a new ISTA/P session must be started, see next page.**

**Enabling is not required to import the Gracenote® DB.**

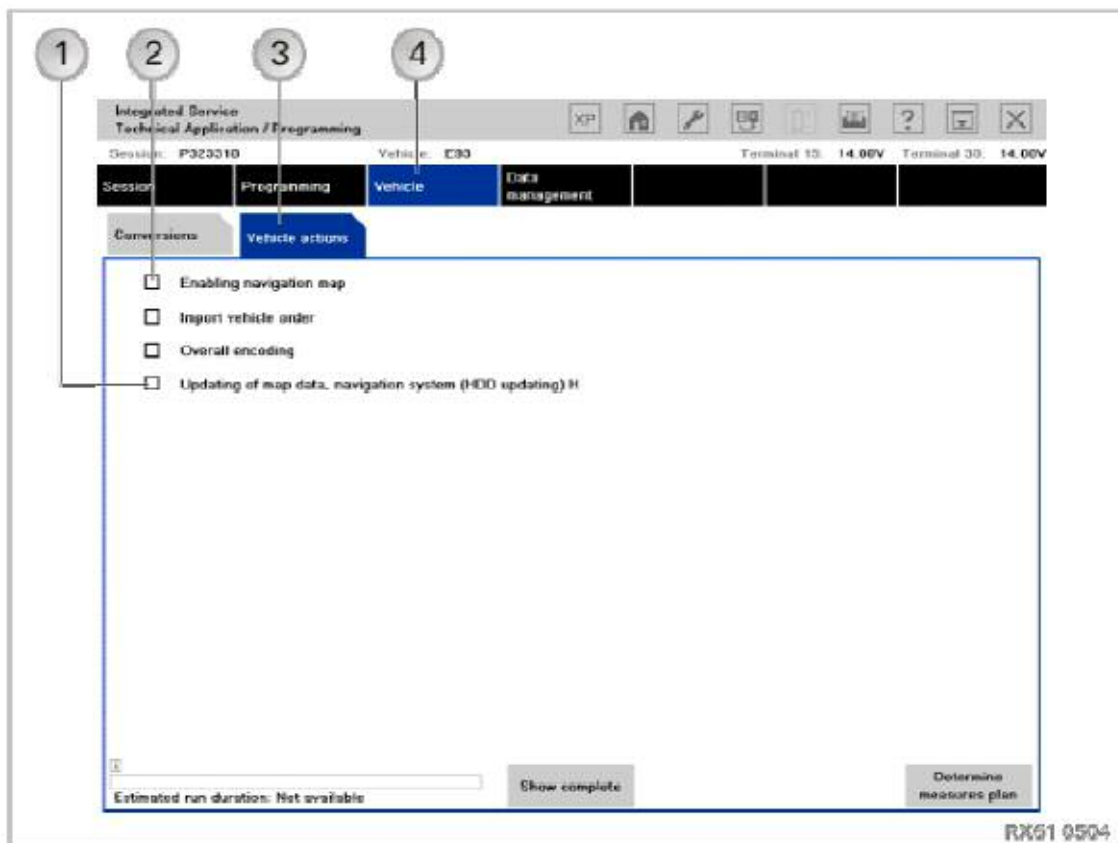
## Enabling of the Navigation Map (Road Map)

An enabling code is required to enable the navigation map. When ordering the enabling code, refer to the section Data management menu.

The enabling code can be imported to the ISTA/P in the following ways:

- Import via Data management before the start of the session.
- Import via SWT online after the action plan has been accepted (as of ISPI 2.7.1).
- Import via external storage medium (e.g.: USB stick, CD) after the action plan has been accepted.

Proceed as follows to activate the map data for the navigation system:



| Index | Screen element                                                                  | Index | Screen element        |
|-------|---------------------------------------------------------------------------------|-------|-----------------------|
| 1     | Checkbox "Update of the navigation system map data (HDD update)"                | 3     | "Vehicle actions" tab |
| 2     | Activate checkbox "Enabling of navigation map (Road Map)", already imported map | 4     | "Vehicle" menu        |

---

To activate a map that has already been imported (e.g. new car):

- Activate the checkbox "Enabling of navigation map".
- Press the "Determine action plan" button to acknowledge.

The remaining procedure depends on the import of the enabling code:

### **Import via Data Management**

The required enabling code is imported from Data management. After the action plan has been completed, the final report will be displayed.

### **Import via SWT Online (as of ISPI 2.7.1)**

The pop-up "Enabling code download via SWT" is displayed.

- Press the "Import" button to acknowledge.

The required enabling code is imported. After the action plan has been completed, the final report will be displayed.

### **Import via External Storage Medium (e.g.: USB stick, CD)**

The pop-up "Enabling code download via SWT" is displayed.

- Press the "Next" button to acknowledge

The pop-up "Enabling code download via USB stick" is displayed.

- Press the "Import" button to acknowledge.

The required enabling code is imported. After the action plan has been completed, the final report will be displayed.

**Note: For further information concerning ISTA/P please refer to the ISTA/P Documents available on the DCS.net under Workshop Technology.**

# Glossary

## A

### **Assembly number**

Hardware number, program version and data version together give the assembly number.

## B

### **Bus systems**

The bus systems enable the individual control units in the vehicle to be networked via serial interfaces. The following bus systems are used in BMW vehicles:

- BSD (Bit-Serial Data interface)
- Byteflight
- CAN bus (Controller Area Network bus)
- DCAN (diagnosis CAN)
- Ethernet (wire-conducted data network technology for local data networks and vehicle access)
- FCAN (suspension CAN)
- FlexRay
- K bus (body bus) (also known as I bus - instrumentation bus - in earlier models)
- K bus protocol
- KCAN (body CAN)
- K-CAN2 (body CAN 2)
- LIN bus (Local Interconnect Network bus)
- Local-CAN
- MOST bus (Media Oriented System Transport)
- PT-CAN (Powertrain CAN)
- PT-CAN2 (Powertrain CAN 2)
- USB (Universal Serial Bus).

---

## C

### **CBS data**

Condition Based Service. CBS data are updated in the vehicle key within a driving cycle. CBS data can be updated on the ignition key via a concealed service function.

### **Car and key memory**

Performs customer-specific settings on the vehicle:

- Car Memory settings affect all the users of a vehicle
- Key Memory settings are user-specific and relate to a key being used.

### **Conversion**

Modifies individual functions in a control unit (e.g. the language); the vehicle must be in sleep mode before complete conversion is carried out.

## D

### **Diagnosis address**

Address of control unit for diagnosis.

## E

### **EWS/DME or EWS/DDE adjustment**

Adjustment between EWS (electronic immobilizer) and DME/DDE control unit, control units are synchronized.

### **Enabling code**

The enabling code generates a functional enhancement in the vehicle. Software enabling in a number of control units (e.g. when programming the CCC, an enabling code must be imported to enable the option "extended voice control). The background to this is the technology of Sweeping Technologies (SWT).

### **Enabling code short**

Comparable to enabling code. In contrast to the enabling code, the enabling code short is not a file. It is used for manual entry purposes.

---

## **Encoding**

Adapts the control units to the vehicle in which they are installed: Functions and characteristic maps are enabled or activated depending on the national-market version, vehicle equipment and vehicle type.

## **F**

### **Flash-programming**

See Programming

## **G**

### **Gracenote® DB**

Music track recognition technology from Gracenote®. It provides the data (metadata) associated with the music tracks.

This metadata contains information such as:

- Artist/performer
- Song ISTA-Programming
- Album
- Year of publication
- Genre

### **Gateway**

This information identifies the four segment address of the component located on the network responsible for communicating from the current network to another network. If there is no address in this location then a connection to any devices outside of the current network can not be established.

## **I**

### **IBAC enabling code**

Enabling code for enabling conversions and retrofits.

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**ICOM**

Integrated Communication Optical Module

The interface between the BMW Workshop network and the vehicle

**IMIB**

Integrated Measurement Interface Box

Measuring device for the ISTA workshop system. The IMIB is mainly operated online via the ISTA workshop system and controlled by the ISID.

**ISAP**

Integrated Service Access Point

Wireless data transmission device for the ITOOLS

**ISID**

Integrated Service Information Display

Display and operating unit for the workshop and vehicle reception

**ISIS**

Integrated Service Information Server (workshop server)

**ISPA**

Integrated Service Processes Application

Software for service reception

**ISSS**

Integrated Software Service Station

By installing the ISTA/P Base DVD via the SSS DVD drive, the SSS becomes an ISSS.

**ISTA**

Integrated Service Technical Application (diagnosis and technical documents)

**ISTA/P**

Integrated Service Application/Programming

(programming system)

**Integration level/Integration stage**

Production periods during vehicle development are denoted by integration levels (I-levels).

**ITOOLS**

The ITOOLS are ISID, ICOM, IMIB and ISAP

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## **IP address**

The IP address (internet protocol address) is a unique and specifically assigned address for each device in the workshop network. This address consists of a number block which is normally structured as follows.

**Example:** 192.168.100.10

The IP addresses must be structured so that each device in the network is distinguished from all the others and must be exclusive to that device and not shared among other devices connected to the network.

## **IP subnet mask**

This information is used to define which segment of the four segment IP address specifically identifies the device and which identify the specific network.

**Example:** 255.255.255.0

Indicates that the first three segments (255.255.255) identify the specific network that the device is located on. The last segment (0) indicates that this is the segment that will identify the specific device.

**Note: The network and device address information must be entered exactly as defined by the network administrator for your facility, otherwise the devices can not be accessed.**

## **J**

### **JETstream**

Online update: Renews application software by loading new software packages.

## **M**

### **MAC address**

Media Access Control address is an unchangeable and unique address for identification of a network device.

### **Mecca indicator**

The "Mecca indicator" is an arrow within the navigation system that points permanently at the geographical position of the city of Mecca.



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## N

### **Native action plan**

Shown after successfully determining target context. All actions are shown that are initiated by ISTA/P (e.g. update integration stage).

Comparable to status report in Progman.

## O

### **Overall encoding**

Matches functions to each other and synchronizes them. This may be necessary if, after programming/encoding is completed, functions in the vehicle fail to work or fail to work properly.

## P

### **Personal profile**

In certain vehicles, this is the new designation for "Car & Key Memory". Settings are made directly in the vehicle. Only those settings which are difficult for the customer to grasp (e.g. tilt sensor ON/OFF) will continue to be made in ISTA/P in the Conversions menu.

### **Programming**

Loads a new program to the control unit. Former designation "flash programming".

### **Programming status**

Indicates the status of the control unit as a number

## R

### **Retrofit**

Designation for a subsequent installation (e.g. telephone), a new system is adapted to the vehicle's complete electrical system.

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**Replacement follow-up**

Follow-up work on a control unit after replacement.

**S****SGC**

Control unit encoding

**SWT**

Sweeping Technology

The enabling code generates a functional enhancement in the vehicle. The background to this is the technology of Sweeping Technologies. The enabling code is encrypted in the control unit.

**Software ID**

Software identification feature

**T****Target context**

Software version of the vehicle which is allocated by the programming system.

The native action plan is shown after successfully determining target context.

**U****USB**

Universal Serial Bus

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## V

### **Vehicle order/central encoding code**

Denotes the file in which various vehicle data (data status, optional equipment, etc.) are kept.

## W

### **Web copy console**

Web copy console for navigation data

The web copy console for navigation data allows you to copy navigation maps for the CIC onto the ISIS hard disks.

### **WSM**

Workshop System Management

Used to manage the ISIS. All administrative tasks are executed here.

## Z

### **ZCS**

Central encoding key, see vehicle order.

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NOTES

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